Abdelaziz Ibrahim

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Profile

AI engineer with a master's degree in Artificial Intelligence, experienced in **LLM applications, model deployment, and evaluation**. Passionate about developing **AI solutions** and designing **agent** workflows. Curious and committed to continuous learning, with a strong academic record at both bachelor's and master's levels.

Professional experience

AI/NLP Researcher, *MaLA team* 10/2025 – 06/2026
Python, LLM, vLLM, Data pipelines, PyTorch, Slurm, MLOps, Docker, Apptainer Remote, Finland

Built the translation **quality estimation** stage of **FineOPUS**, which prepares massively multilingual OPUS parallel data for LLM and machine translation training. Master's thesis contribution: evaluator benchmarking, model selection, and scalable scoring/filtering.

- Unified incompatible evaluators behind five backends: **COMET, MetricX-24, ReMedy, Bicleaner-AI**, and **vLLM** judges (e.g., **Qwen**); normalized language codes across models and datasets for consistent language coverage comparisons.
- Benchmarked **nine evaluators** across **41,412 translation directions**, tested and analyzed unsupervised ensembles. Different winners by direction-level, overall average score, and rank stability motivated **per-direction model routing**.
- Analyzed documented **language coverage** of each model: target-only coverage was associated with higher QE scores than source-only coverage across all nine models, informing **coverage-aware evaluator selection**.
- Handled malformed LLM responses with **Pydantic/JSON schemas**, item-ID/range validation, and **selective retries**, producing validated records for automated scoring.
- Resolved AMD GPU dependency conflicts by source-building **vLLM for ROCm** in **Apptainer**, matching PyTorch and AMD SMI to ROCm and excluding CUDA wheels to enable LLM scoring on **LUMI MI250X**.
- Managed GPU memory with **length-sorted batching, bfloat16**, and context limits for **Slurm** jobs on Mahti A100 and LUMI MI250X.
- Used preassigned manifests for jobs to ensure **atomic writes**, added **resumable checkpoints**, shard consolidation, and recovery tests for interrupted jobs.
- Applied benchmark-driven routing with streaming **Parquet/DuckDB** processing; scored **1.14 billion pairs** in a completed 421-direction subset, retaining **63.43%** at a 0.7 quality threshold.
- Co-authored the Model-Based Quality Assessment paper; released pipeline code and filtered data.

Teaching Assistant, *Helwan University* 06/2022 – 08/2024
NLP, Machine learning, Evolutionary algorithms, Data structures, Algorithms Helwan, Egypt

- Taught **NLP, machine learning**, and **evolutionary algorithms** through theory and implementation examples.
- Guided **data preprocessing, model evaluation**, and **code debugging** to help students interpret results.
- Prepared **teaching materials and examinations**; graded work and gave feedback on correctness and reasoning.

Database Specialist, *Armed Forces (military service)* 09/2021 – 06/2022
Oracle SQL, .NET, Technical support Cairo, Egypt

- Built **Oracle SQL** reports on personnel counts, service/medical status, conflict veterans, and families using joins, aggregations, and conditional filters.
- Traced faults through **.NET** application logic and database queries; reproduced errors and revised queries to improve reporting efficiency and restore workflows.
- Managed organization's website, fixed data-retrieval defects, providing **technical support** for personnel systems.

Education

Master's Degree in Artificial Intelligence *(GPA: 4.89/5.0)*, *University of Jyväskylä* 08/2024 – 06/2026
Coursework: LLMs, Agentic AI, cloud computing, and knowledge graphs. Jyväskylä, Finland

Bachelor of Computer Science *(Top of class; GPA: 3.83/4.0)*, *Helwan University* 09/2016 – 05/2020
Coursework: Software Engineering, Data Structures, and Information Systems. Helwan, Egypt

Selected Projects

SmartBi (Norrin) challenge winner **AaltoAI Hackathon 2025**

1) SQL data analytics agent

LangGraph, tool call, Flask, SQLite, Chart.js

Agent that turns **natural-language questions** into database queries and **visualizes the results**.

- Enabled database exploration without writing SQL, built a **LangGraph ReAct** agent with **SQLDatabaseToolkit** for schema discovery, query execution, and prompted error correction within a bounded tool loop.
- Made complex query outputs usable by the frontend: parsed **CTEs, aliases, and nested expressions** into named records and exposed SQL, results, and schema through **Flask APIs**.
- Handled inconsistent and duplicate chart output by separating insight/chart generation, validating types and datasets, and deduplicating with **content fingerprints**; returned **Chart.js** visualizations and contextual summaries.
- Designed **query expansion prompts** that use the original question, generated SQL, and query results to suggest related analyses and secondary charts. Stored each secondary chart's corresponding question in its metadata, allowing users to select a chart and submit the prefilled question for **SQL generation and database execution**.

Projects (continued)

2) Legal document processing (Lawgic) *FastAPI, RAG, OCR, Chroma, Pydantic, SQLAlchemy*

Legal-document processing with OCR, conversational RAG, structured contract extraction, translation, and EU law search.

- Reconciled mismatched OCR text and document structure by combining **olmOCR** with **Granite-Docling layout analysis**. Used **RapidFuzz**, word overlap, and confidence thresholds to align text and replace matched regions while preserving layout metadata.
- Kept long-running OCR outside the request cycle through **FastAPI background tasks**, status polling, timeouts, and **SQLAlchemy** state tracking, allowing uploads to progress asynchronously to searchable documents.
- Preserved contract sections for retrieval with **header-aware chunking**, Hugging Face embeddings, and source metadata; used **incremental Chroma indexing** and **SQLRecordManager** to update the document index without accumulating duplicate chunks.
- Resolved ambiguous follow-up questions through **history-aware query rewriting**; used **MMR retrieval** to diversify passages and **long-context reordering** to organize evidence, then streamed answers and saved conversation/source history.
- Turned unstructured clauses into **contract entities, relationships, dates, and terms** using chunk-wise LLM extraction and **Pydantic schemas** with JSON fallback; deduplicated parties/relationships and persisted them with **SQLAlchemy transactions**.
- Connected contract content to EU legislation by generating search keywords and querying **EUR-Lex SOAP/XML**; applied language and legislation filters and merged duplicate results into a focused research response.
- Supported multilingual review with **page-ordered translation**, thread-pool execution, and document/language caching to reuse completed results; enforced **JWT authentication**, ownership, and OCR-readiness checks on translation requests.

Languages

English: Fluent **Finnish:** Beginner **Arabic:** Native

Technical skills

Programming languages: Python, SQL
AI and inference: PyTorch, Transformers, vLLM
Databases: Oracle SQL, PostgreSQL, DuckDB, SQLite
Knowledge graphs: SPARQL, RDF
Containers: Docker, Apptainer
Applications and APIs: FastAPI, Flask, Streamlit
Agents and evaluation: DSPy, LangChain, LangGraph, Langfuse
Data processing: Parquet, PySpark
HPC and GPUs: Slurm (HPC), CUDA/ROCm environments
Cloud: AWS fundamentals

Soft skills

• **Problem-solving** • **Adaptability** • **Leadership and mentoring** • **Collaboration and teamwork** • **Curiosity and perseverance**